

Tahmid Zalal (Tim)

St. John's, NL | tahmidmahin89@gmail.com | (709) 330-8336 | LinkedIn: [linkedin.com/in/tahmidzalal](https://www.linkedin.com/in/tahmidzalal) | GitHub: [xtimmy888](https://github.com/xtimmy888)

PROFESSIONAL SUMMARY

Computer Science and Physics graduate with experience in scientific computing, machine learning, and data analytics. Skilled in developing computational workflows, building predictive models, and analyzing complex datasets through research and software development. Strong analytical, problem-solving, and communication skills.

EDUCATION

Memorial University of Newfoundland

Joint Bachelor of Science in Computer Science and Physics

Graduated: April 2026

St. John's, NL

Relevant Coursework: Data Visualization; Data Mining; Machine Learning; Data Structures & Algorithms

WORK EXPERIENCE

Software Developer

Sep 2025 – Present

MUN Quant Society Data & Infrastructure Team (Student Society)

St. John's, NL

- Contributed to and maintained MQS Data & Infrastructure tooling, supporting sentiment pipelines for news/social ingestion, NLP scoring, storage, and downstream analytics using web APIs, pandas, and text-cleaning.
- Generated per-ticker daily sentiment scores and structured financial/market datasets for faster analysis, visualization, and strategy testing.
- Implemented and validated a Python backtesting engine using pandas and NumPy for signal generation, portfolio weights, risk controls, transaction costs, and performance reporting across multiple equity tickers.
- Improved reliability and reproducibility through GitHub Actions, Docker, and SQL/PostgreSQL time-series storage, making strategies plug-and-play and reducing setup time from 1–2 hours to under 1 hour.

Computational Chemistry Research Assistant

Jan 2026 – Present

Memorial University of Newfoundland (Department of Chemistry)

St. John's, NL

- Built and ran an all-atom GROMACS molecular dynamics workflow for ubiquitin PDB 1UBQ on Narval HPC, supporting computational chemistry research with a CHARMM36/TIP3P protein–water–ion system.
- Completed the full simulation pipeline, including system preparation, energy minimization, NVT/NPT equilibration, production MD, Slurm-based HPC jobs, and checkpoint recovery.
- Processed trajectory outputs using Python, NumPy, pandas, Matplotlib/Seaborn, and GROMACS tools, analyzing RMSD, RMSF, and H-bond stability trends across 251 simulation frames to support computational research analysis.

Scientific Computing Research Assistant

Jan 2026 – May 2026

Memorial University of Newfoundland (Department of Mathematics and Statistics)

St. John's, NL

- Built and validated a Python/Fortran adaptive RK45/Dormand–Prince 5(4) ODE solver workflow, generating 1,115,989 adaptive step attempts across 7,810 runs on scalar and coupled first-order ODE systems, including exponential decay, logistic growth, SHO, VDP, and Lorenz dynamics.
- Engineered a leakage-free ML dataset from solver telemetry using run-level train/validation/test splits and pre-step features such as step size, tolerance, state norm, slope norm, ODE parameters, and problem type to predict likely rejected adaptive steps without using post-step error information.
- Trained and evaluated Logistic Regression and XGBoost reject-prediction models, with the tuned XGBoost model achieving 0.916 PR-AUC, 90.7% reject precision, 73.1% reject recall, and 0.810 reject F1; compiled methods, results, and solver-policy findings into a structured research paper draft.

SKILLS

Programming & Development: Python, Java, JavaScript, HTML, CSS, SQL, Node.js, React.js, C, Fortran | Data & ML: pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn, Plotly, GROMACS | Tools & Platforms: Git, GitHub, Jupyter Notebook, Linux, PostgreSQL, Docker, GitHub Actions, Slurm, Narval HPC.